

22

WEDNESDAY

22/10/2025

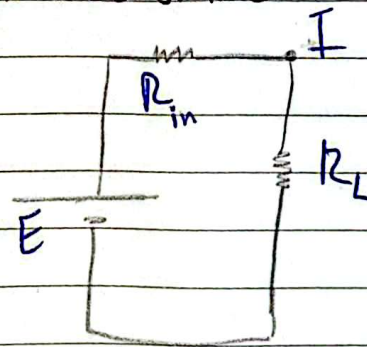
الأربعاء

Maximum Power Transfer

A resistive load will abstract maximum Power from a network when the load resistance is equal to the resistance of the network as viewed from the output terminals with all source of emf. removed behind their internal resistance

$$P = I^2 R_L$$

$$I = \frac{E}{R_i + R_L}$$



$$P = \left(\frac{E}{R_i + R_L} \right)^2 R_L$$

P_{max} When $R_i = R_L$

$$P_{max} = \left(\frac{E}{R_L + R_i} \right)^2 R_L = \left(\frac{E}{2R_L} \right)^2 R_L$$

$$P_{max} = \frac{E^2}{4R_L} \Rightarrow P_{max} = \frac{E^2}{4R_L}$$

Week 43

NOTES

ملاحظات

تشرين ثاني November 11

Sat.	29	22	15	08	01	السبت
Sun.	30	23	16	09	02	الأحد
Mon.		24	17	10	03	الاثنين
Tue.		25	18	11	04	الثلاثاء
Wed.		26	19	12	05	الأربعاء
Thu.		27	20	13	06	الخميس
Fri.		28	21	14	07	الجمعة

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TUESDAY

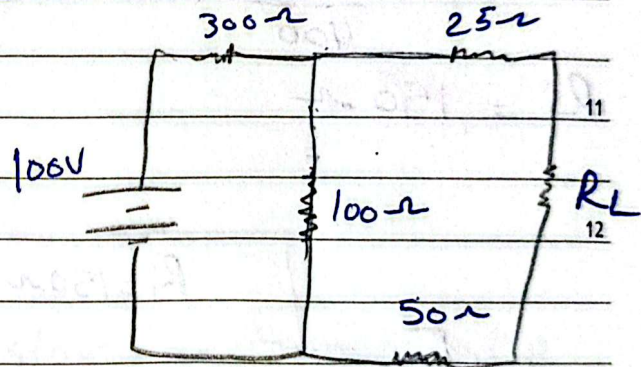
21/10/2025

الثلاثاء

Exempl (1)

Find the value of R_L necessary to obtain maximum Power in R_L .
Hence evaluate the maximum Power dissipated at R_L .

$$P_{\max} = \frac{E^2}{4R_L}$$



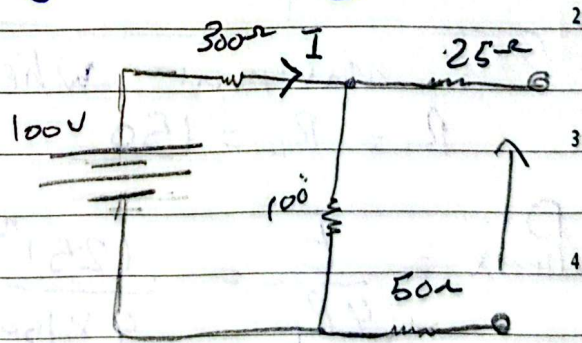
Remove R_L from network and calculate open circle voltage E (Apply KVL)

$$I = \frac{100}{300 + 100} = \frac{100}{400}$$

$$I = 0.25 \text{ A}$$

$$E = I \times 100 = 0.25 \times 100$$

$$E = 25$$



الأسبوع ٤٣

تشرين أول October 10

NOTES

ملاحظات

Sat.	25	18	11	04	السبت
Sun.	26	19	12	05	الأحد
Mon.	27	20	13	06	الاثنين
Tue.	28	21	14	07	الثلاثاء

20

MONDAY

20/10/2025

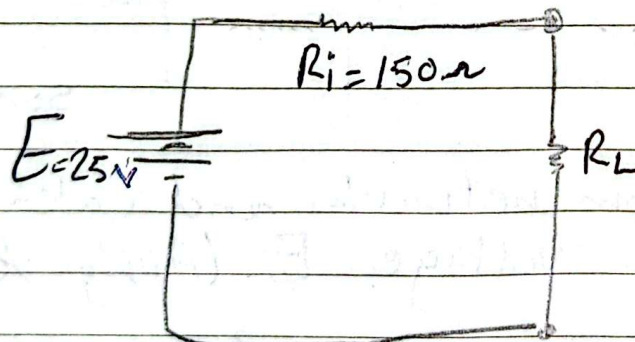
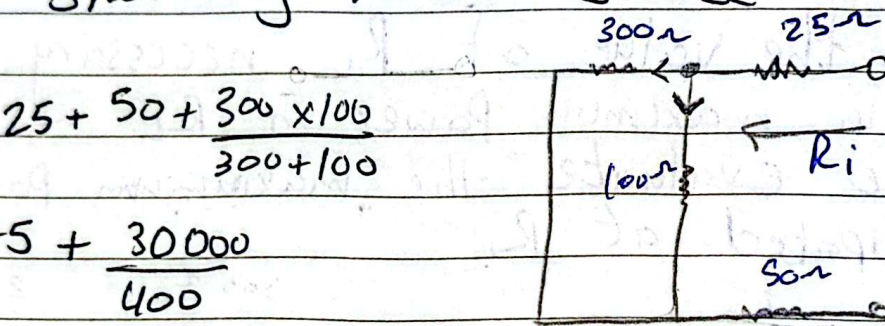
الاثنين

By shorting 100V Source

$$R_i = 25 + 50 + \frac{300 \times 100}{300 + 100}$$

$$R_i = 75 + \frac{30000}{400}$$

$$R_i = 150 \Omega$$



P is maximum when
 $R_i = R_L = 150$

$$P_{max} = \frac{E^2}{4R_L} = \frac{(25)^2}{4 \times 150}$$

$$P_{max} = 1.042 \text{ watt}$$

Week 43

NOTES

ملاحظات

تشرين ثاني November 11

السبت	01	08	15	22	29	Sat.
الأحد	02 <td>09 <td>16 <td>23 <td>30 <td></td> </td></td></td></td>	09 <td>16 <td>23 <td>30 <td></td> </td></td></td>	16 <td>23 <td>30 <td></td> </td></td>	23 <td>30 <td></td> </td>	30 <td></td>	

19

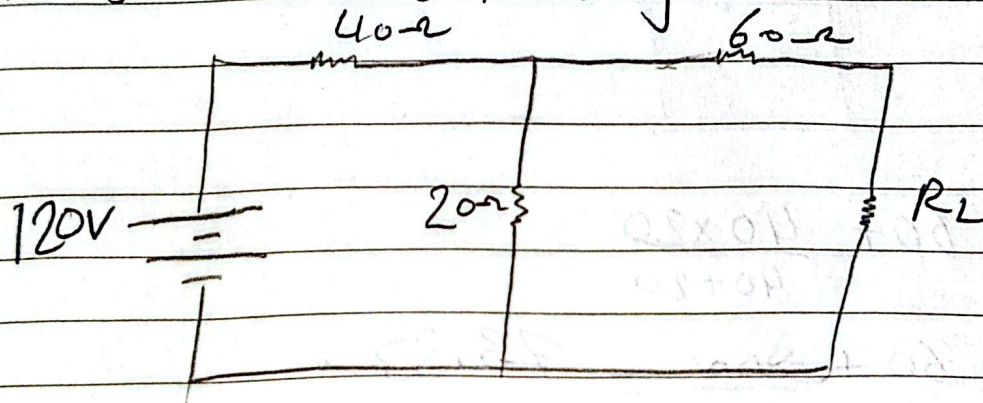
SUNDAY

19/10/2025

الأحد

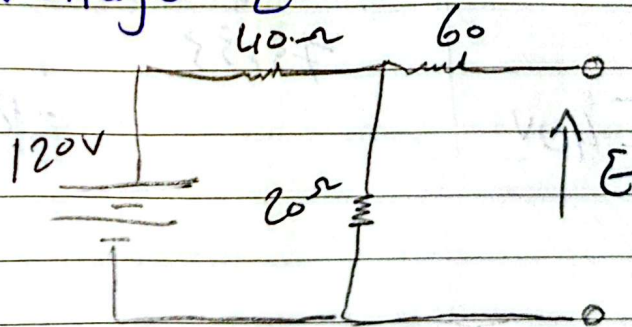
Example (2)

Find the maximum Power for R_L resistor of the following circuit



Remove R_L resistor and calculate open circuit voltage E

$$I = \frac{120}{40 + 20} = 2A$$



$$E = I \times 20$$

$$E = 2 \times 20 = 40V$$

الأسبوع ٤٣

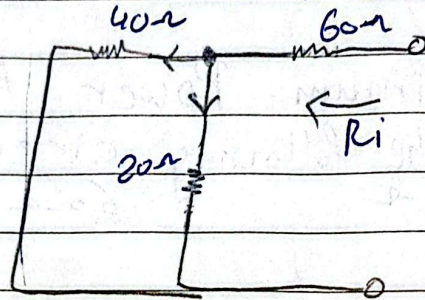
تشرين أول October 10

NOTES

ملاحظات

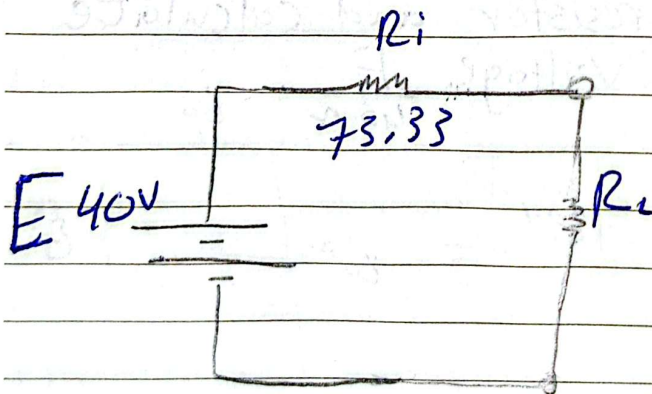
Sat.	25	18	11	04	السبت
Sun.	26	19	12	05	الأحد

Shorting 120V source & calculate R_i



$$R_i = 60 + \frac{40 \times 20}{40 + 20}$$

$$R_i = 60 + \frac{800}{60} = 73.33 \Omega$$



Maximum Power will be when

$$R_i = R_L = 73.33 \Omega$$

$$P_{\max} = \frac{E^2}{4R_L} = \frac{(40)^2}{4 \times 73.33} = 5.45 \text{ watt}$$

Week 43

NOTES

ملاحظات

تشرين ثاني November 11

Sat.	29	22	15	08	01	السبت
Sun.	30	23	16	09	02	الأحد
Mon.		24	17	10	03	الاثنين
Tue.		25	18	11	04	الثلاثاء
Wed.		26	19	12	05	الأربعاء
Thu.						

16

THURSDAY

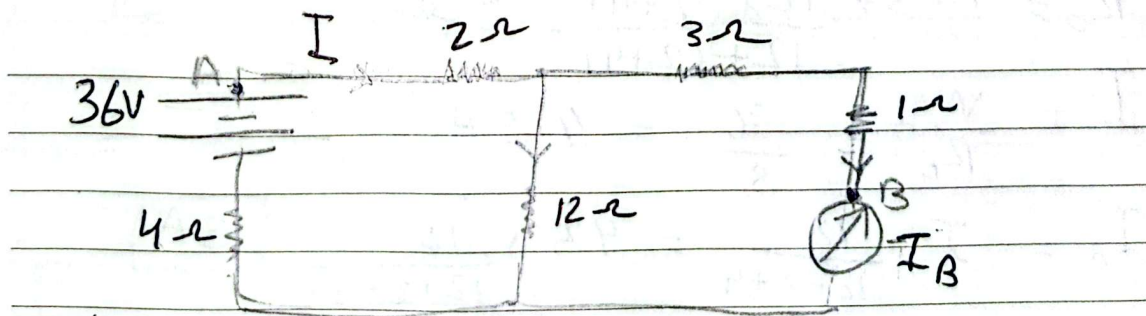
16/10/2025

الخميس

Reciprocity theorem

In any linear bilateral network, if a source of emf E in any branch produces current I in any other branch, then the zero emf E acting in the second branch would produce the same current I in the first branch.

Example (1) : verify reciprocity theorem



الأسبوع ٤٢

October 10 تشرين أول					
Sat.	25	18	11	04	السبت
Sun.	26	19	12	05	الأحد
Mon.	27	20	13	06	الاثنين
Tue.	28	21	14	07	الثلاثاء
Wed.	29	22	15	08	الأربعاء
Thu.	30	23	16	09	الخميس
Fri.	31	24	17	10	الجمعة

NOTES

ملاحظات

15

WEDNESDAY

15/10/2025

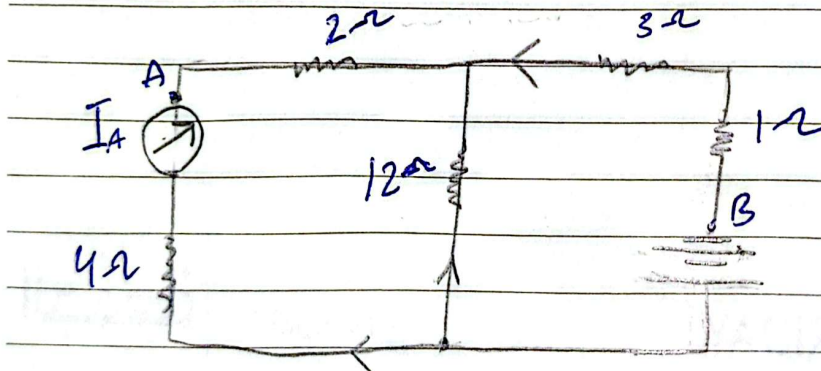
الأربعاء

$$R_t = 4 + 2 + \frac{12 \times (3 + 1)}{12 + (3 + 1)}$$

$$R_t = 9 \Omega$$

$$I = \frac{V}{R} = \frac{36}{9} = 4A$$

$$I_B = I \cdot \frac{12}{12 + 3 + 1} = 4 \times \frac{12}{16} = \boxed{3A}$$



$$R_t = 1 + 3 + \frac{12 \times (2 + 4)}{12 + (2 + 4)} = 8 \Omega$$

$$I = \frac{V}{R_t} = \frac{36}{8} = 4.5A$$

$$I_A = I \cdot \frac{12}{12 + 2 + 4} = 4.5 \times \frac{12}{18} = \boxed{3A}$$

Week 42

NOTES

ملاحظات

تشرين ثاني November 11

Sat.	29	22	15	08	01	السبت
Sun.	30	23	16	09	02	الأحد
Mon.		24	17	10	03	الاثنين
Tue.		25	18	11	04	الثلاثاء
Wed.		26	19	12	05	الأربعاء
Thu.		27	20	13	06	الخميس
Fri.		28	21	14	07	الجمعة

14

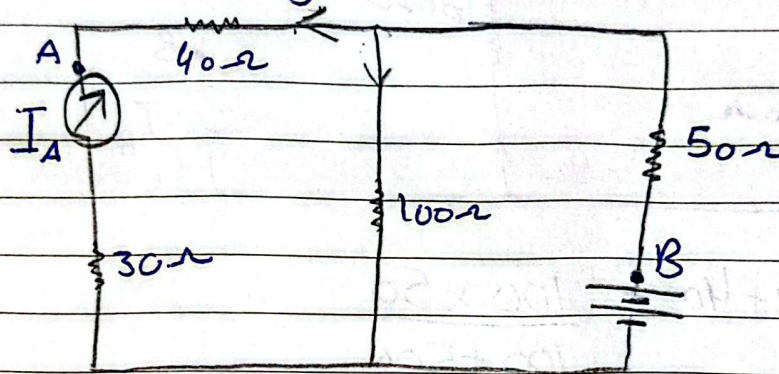
TUESDAY

14/10/2025

الثلاثاء

Example (2)

verify reciprocity theorem to measure the current by Ammeter



$$R_t = 50 + \frac{100 \times (30 + 40)}{100 + (30 + 40)}$$

$$R_t = 50 + \frac{7000}{170} = 91.17 \Omega$$

$$I = \frac{V}{R_t} = \frac{25}{91.17} = 0.274 \text{ A}$$

$$I_A = I \cdot \frac{100}{100 + 40 + 30} = 0.274 \times \frac{100}{170}$$

$$I_A = 0.274 \times \frac{100}{170} = 0.161 \text{ A}$$

الأسبوع ٤٢

تشرين أول October 10

Sat.	25	18	04	السبت
Sun.	26	19	05	الأحد
Mon.	27	20	06	الاثنين
Tue.	28	21	07	الثلاثاء
Wed.	29	22	08	الأربعاء
Thu.	30	23	09	الخميس
Fri.	31		10	الجمعة

NOTES

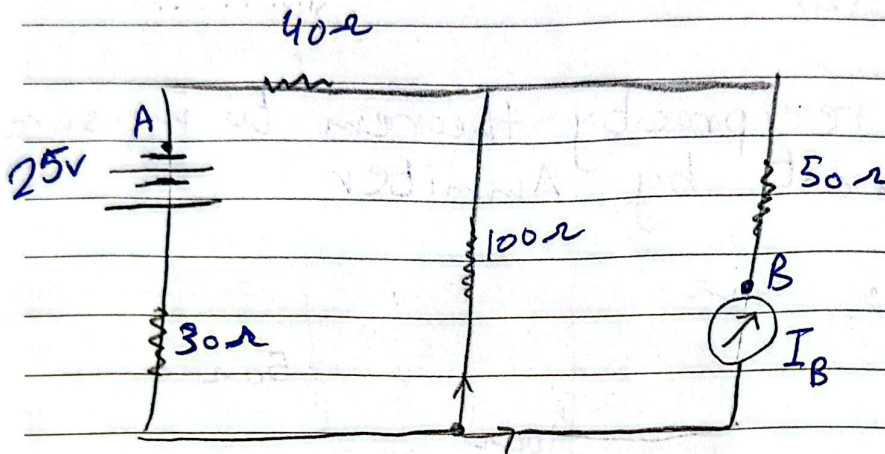
ملاحظات

13

MONDAY

13/10/2025

الاثنين



$$R_t = 30 + 40 + \frac{100 \times 50}{100 + 50}$$

$$R_t = 70 + \frac{5000}{150} = 103.33 \Omega$$

$$I_t = \frac{V}{R_t} = \frac{25}{103.33} = 0.241 \text{ A}$$

$$I_B = I \cdot \frac{100}{100 + 50} = 0.241 \times \frac{100}{150}$$

$$I_B = 0.161 \text{ A}$$

Week 42

NOTES

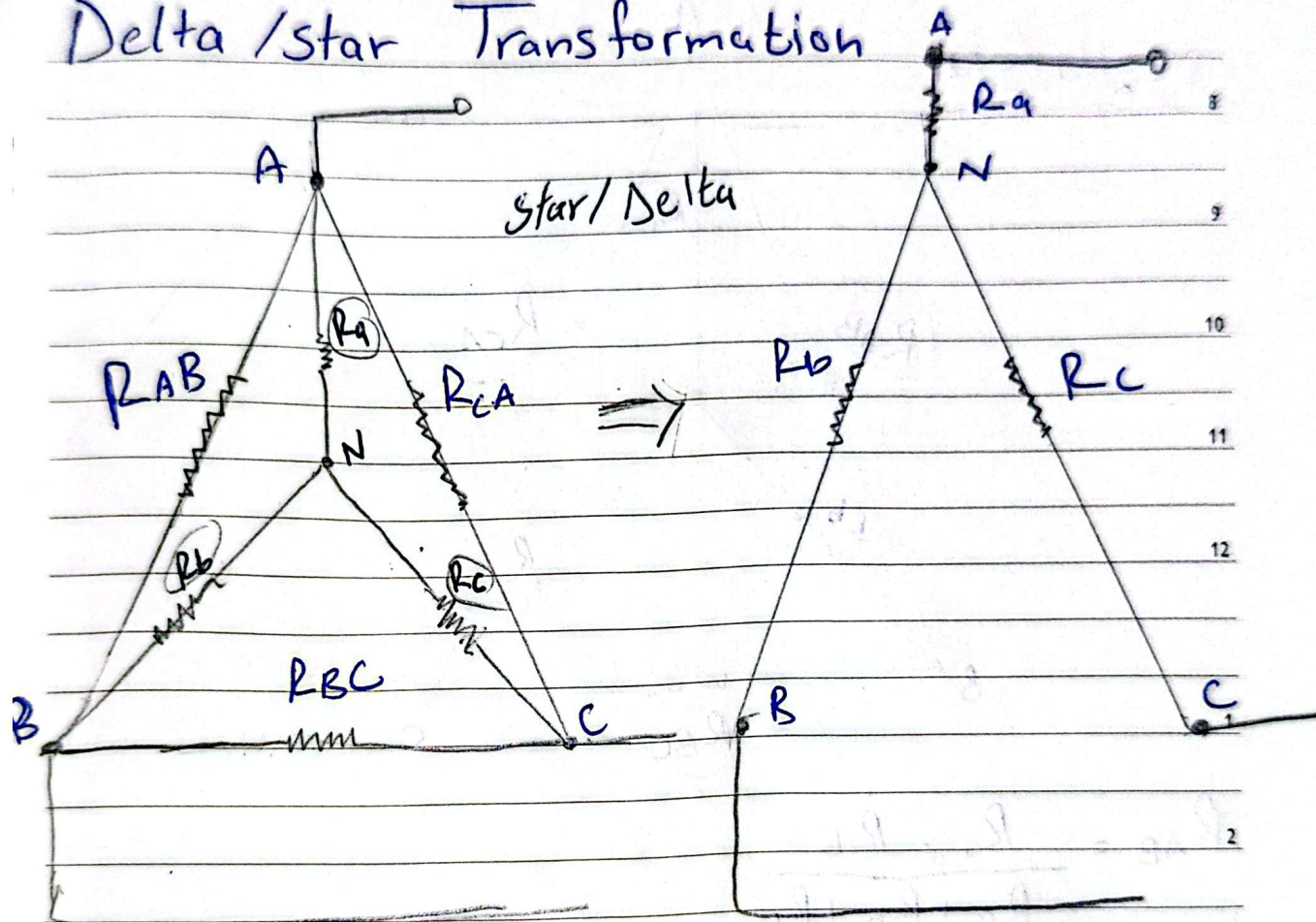
ملاحظات

تشرين ثاني November 11

Sat.	29	22	15	08	01	السبت
Sun.	30	23	16	09	02	الأحد
Mon.		24	17	10	03	الاثنين
Tue.		25	18	11	04	الثلاثاء
Wed.		26	19	12	05	الأربعاء
Thu.		27	20	13	06	الخميس

48

Delta / star Transformation



$$R_a = R_{AB} + R_{CA} + \frac{R_{AB} R_{CA}}{R_{BC}}$$

$$R_b = R_{AB} + R_{BC} + \frac{R_{AB} R_{BC}}{R_{CA}}$$

$$R_c = R_{BC} + R_{CA} + \frac{R_{BC} R_{CA}}{R_{AB}}$$

الأسبوع ٥٢

October 10 تشرين أول

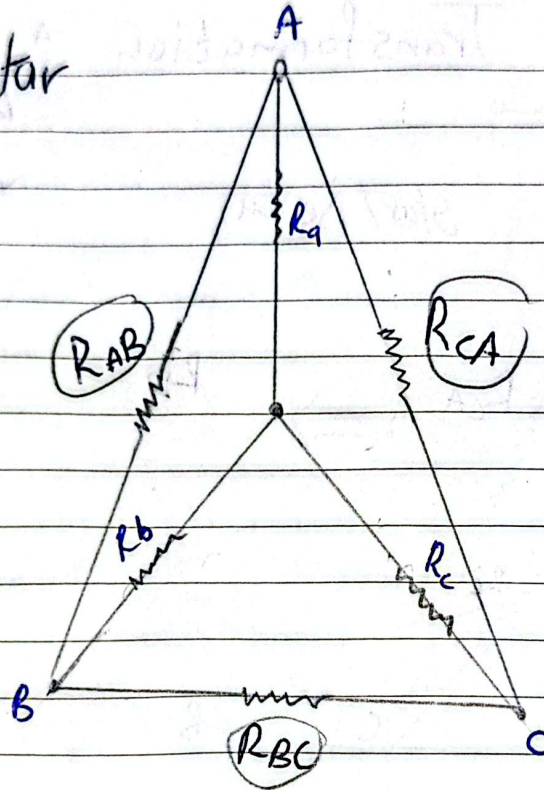
Sat.	25	18	11	04	السبت
Sun.	26	19	12	05	الأحد
Mon.	27	20	13	06	الاثنين
Tue.	28	21	14	07	الثلاثاء
Wed.	29	--	--	--	الأربعاء

NOTES

ملاحظات

from delta to a star

Delta/star



$$R_{AB} = \frac{R_a \cdot R_b}{R_a + R_b + R_c}$$

$$R_{BC} = \frac{R_b \cdot R_c}{R_a + R_b + R_c}$$

$$R_{CA} = \frac{R_a \cdot R_c}{R_a + R_b + R_c}$$

Week 42

NOTES

ملاحظات

from a star to delta

تشرين ثاني November 11

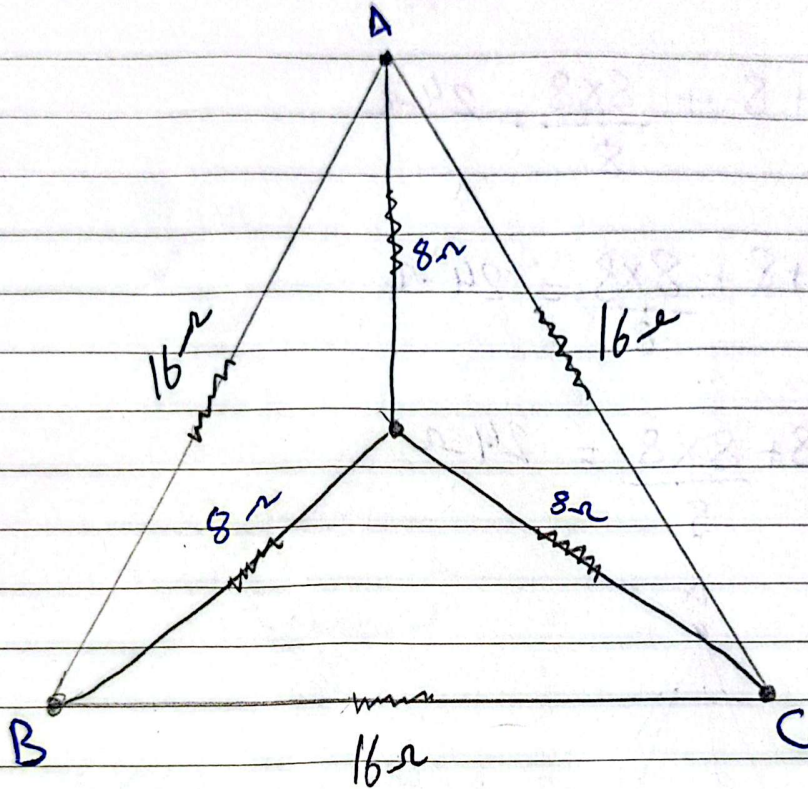
Sat.	29	22	15	08	01	السبت
Sun.	30	23	16	09	02	الأحد
Mon.		24	17	10	03	الاثنين
Tue.		25	18	11	04	الثلاثاء
Wed.		26	19	12	05	الأربعاء
Thu.		27	20	13	06	الخميس

9

THURSDAY

9/10/2025

الخميس



10

FRIDAY

10/10/2025

الجمعة

Evaluate total resistance between A and B using delta/star transformation.

الأسبوع ٤١

تشرين أول October 10

Sat.	25	18	11	04	السادس
Sun.	26	19	12	05	الأحد
Mon.	27	20	13	06	الاثنين
Tue.	28	21	14	07	الثلاثاء
Wed.	29	22	15	08	الأربعاء
Thu.					

NOTES

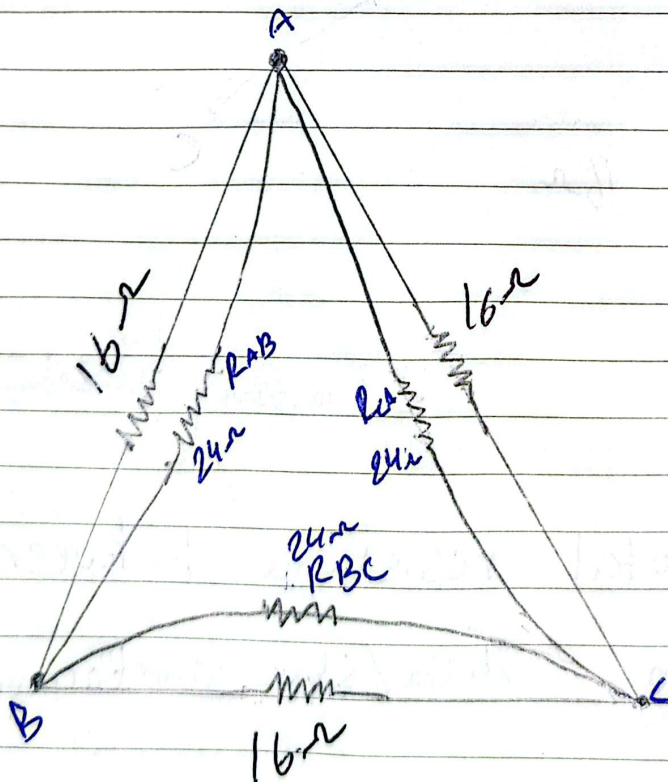
ملاحظات

cuz here it a star we will transform it to delta

$$R_{AB} = 8 + 8 + \frac{8 \times 8}{8} = 24 \Omega$$

$$R_{BC} = 8 + 8 + \frac{8 \times 8}{8} = 24 \Omega$$

$$R_{CA} = 8 + 8 + \frac{8 \times 8}{8} = 24 \Omega$$



Week 41

NOTES

ملاحظات

تشرين ثاني November 11

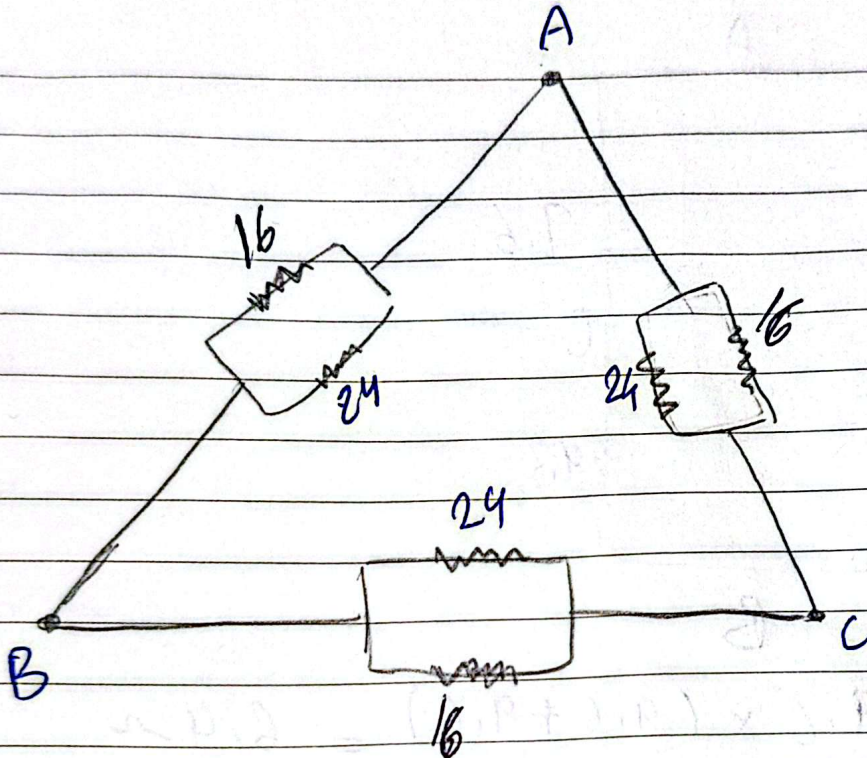
Sat.	29	22	15	08	01	الجمعة
Sun.	30	23	16	09	02	الأحد
Mon.		24	17	10	03	الاثنين
Tue.		25	18	11	04	الثلاثاء
Wed.		26	19	12	05	الأربعاء

7

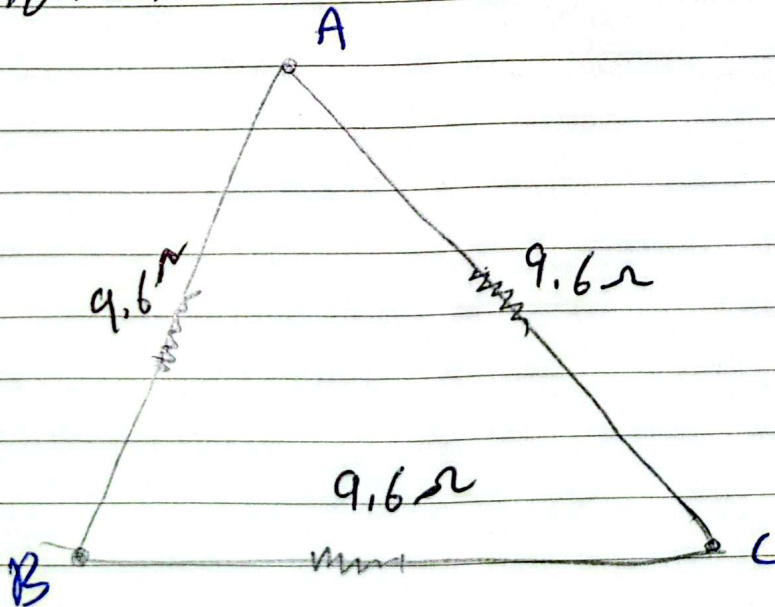
TUESDAY

7/10/2025

الثلاثاء



$$R = \frac{16 \times 24}{16 + 24} = 9.6 \Omega$$



الأسبوع ٤١

تشرين أول October 10

NOTES

ملاحظات

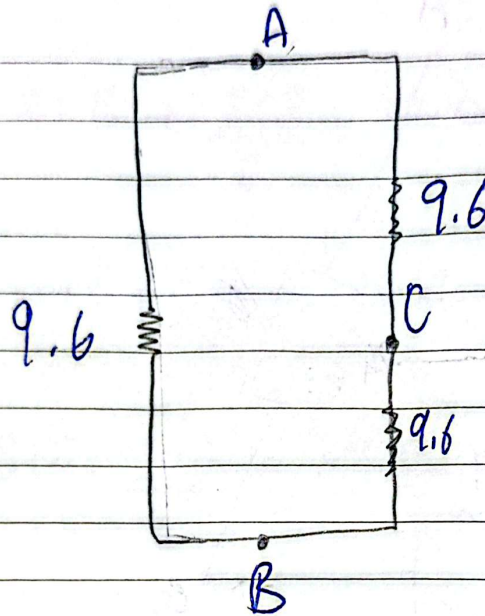
Sat.	25	18	11	04	السبت
Sun.	26	19	12	05	الأحد
Mon					

6

MONDAY

6/10/2025

الاثنين



$$R_{a \text{ and } b} = \frac{9.6 \times (9.6 + 9.6)}{9.6 + (9.6 + 9.6)} = 6.4 \Omega$$

Week 41

NOTES

ملاحظات

تشرين ثاني November 11

Sat.	29	22	15	08	01	السبت
Sun.	30	23	16	09	02	الأحد
Mon.		24	17	10	03	الاثنين
Tue.		25	18	11	04	الثلاثاء
Wed.		26	19	12	05	الأربعاء
Thu.		27	20	13	06	الخميس
Fri.		28	21	14	07	الجمعة

5

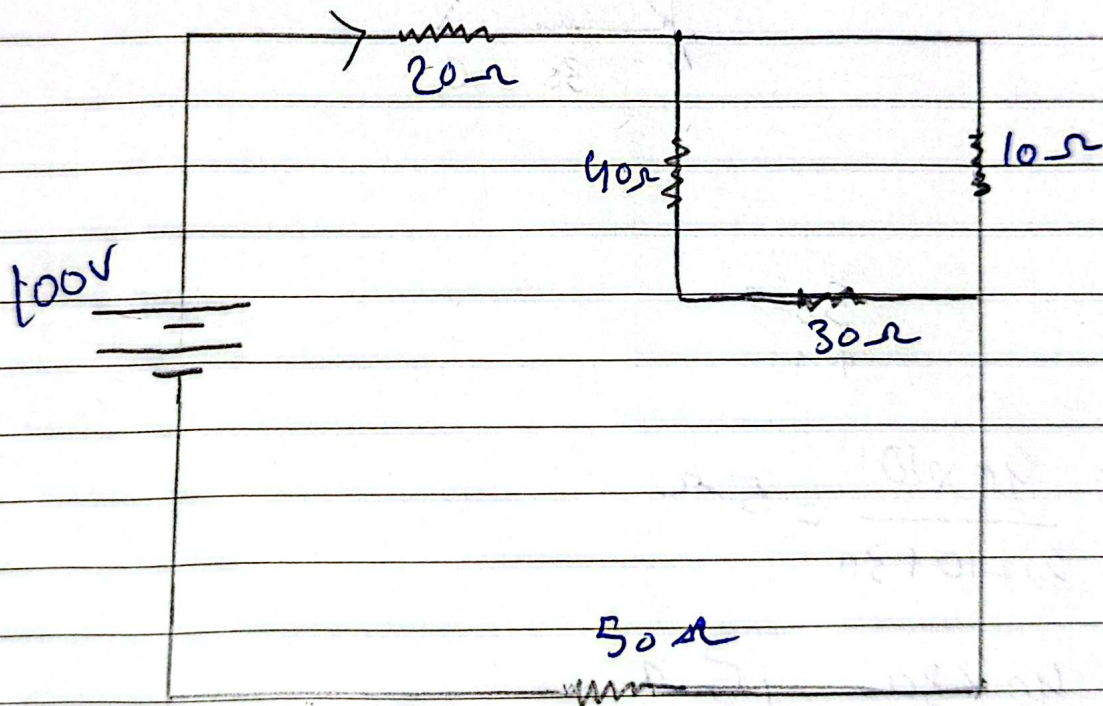
SUNDAY

5/10/2025

الأحد

Example (e)

calculate the total current and total Power of the given circuit

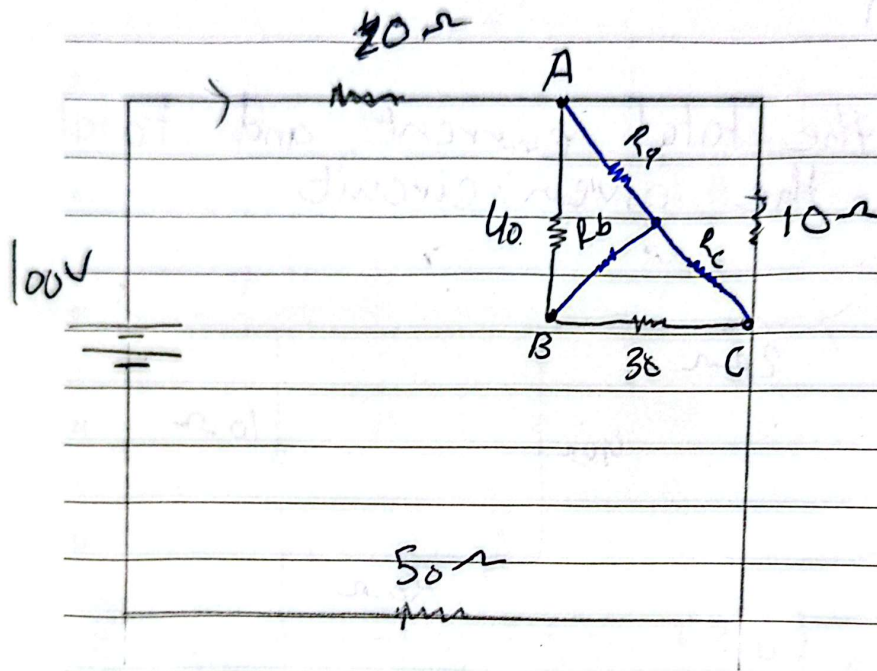


4

SATURDAY

4/10/2025

السبت



$$R_a = \frac{40 \times 10}{40 + 10 + 30} = 5 \Omega$$

$$R_b = \frac{40 \times 30}{40 + 10 + 30} = 15 \Omega$$

$$R_c = \frac{30 \times 10}{30 + 10 + 40} = 3.75 \Omega$$

Week 4

NOTES

ملاحظات

تشرين ثاني November 11

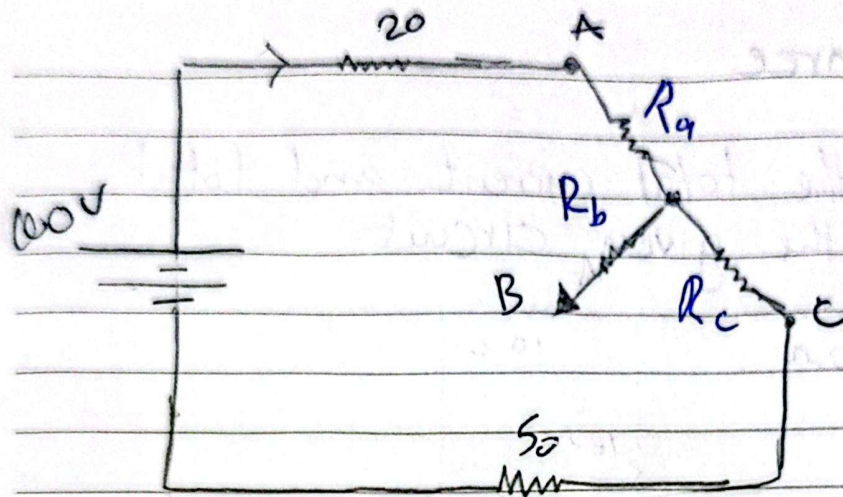
Sat.	29	22	15	08	01	الجمعة
Sun.	30	23	16	09	02	الاحد
Mon.		24	17	10	03	الاثنين
Tue.		25	18	11	04	الاثنين
Wed.		26	19	12	05	الاثنين
Thu.		27	20	13	06	الاثنين

2

THURSDAY

2/10/2025

الخميس



$$R_t = 20 + R_a + R_c + 50$$

$$R_t = 20 + 5 + 3.75 + 50 = 78.75 \Omega$$

3

FRIDAY

3/10/2025

الجمعة

$$I = \frac{V}{R_t} = \frac{100}{78.75} = 1.27 \text{ Ampere}$$

$$P = VI = 100 \times 1.27 = 127 \text{ watt}$$

الأسبوع ٤٠

تشرين أول October 10						
Sat.	25	18	11	04		السبت
Sun.	26	19	12	05		الأحد
Mon.	27	20	13	06		الاثنين
Tue.	28	21	14	07		الثلاثاء
Wed.	29	22	15	08	01	الأربعاء

NOTES

ملاحظات

B is open that why we didn't calculate it

1

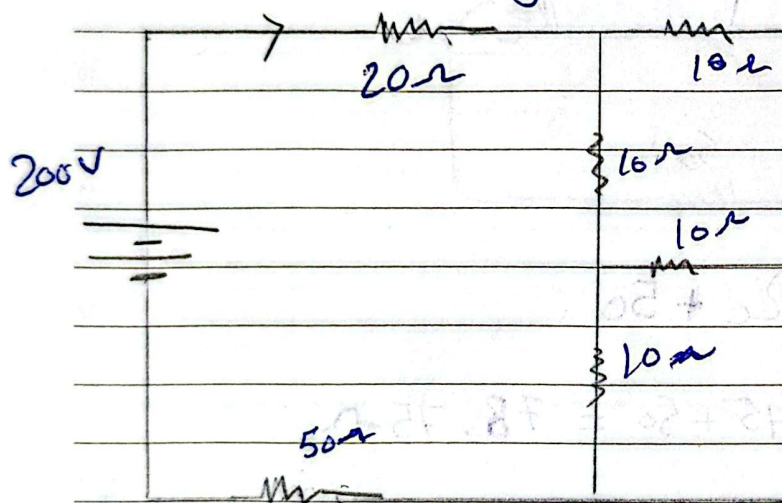
WEDNESDAY

1/10/2025

الأربعاء

Example (3)

Calculate the total current and total Power of the given circuit



$$R_a = \frac{10 \times 10}{10 + 10 + 10} = \frac{100}{30} = 3.33 \Omega$$

$$R_b = \frac{10 \times 10}{10 + 10 + 10} = 3.33 \Omega$$

$$R_c = \frac{10 \times 10}{10 + 10 + 10} = 3.33 \Omega$$

Week 40

NOTES

ملاحظات

تشرين أول October 10

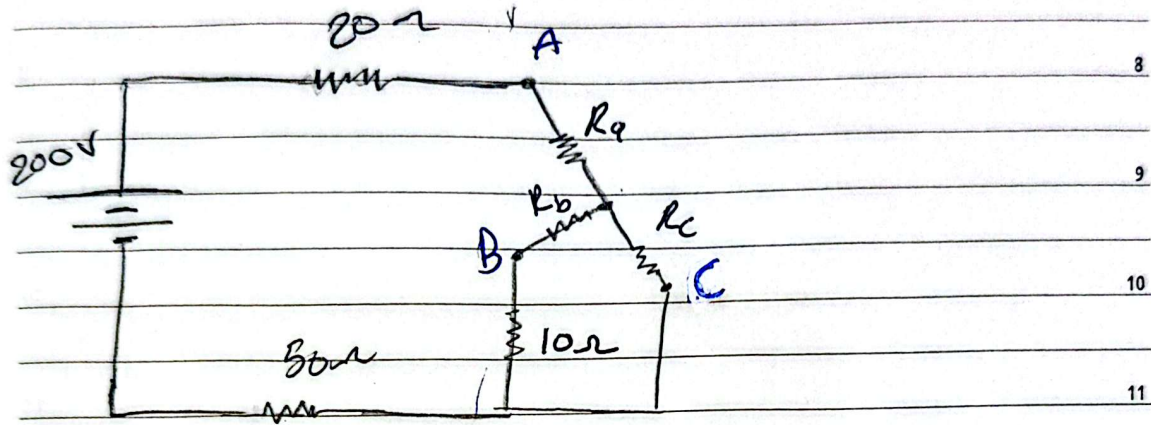
Sat.	25	18	11	04	السبت
Sun.	26	19	12	05	الأحد
Mon.	27	20	13	06	الاثنين
Tue.	28	21	14	07	الثلاثاء
Wed.	29	22	15	08	الأربعاء

30

TUESDAY

30/9/2025

الثلاثاء



$$R_t = 20 + R_q + \frac{(R_b + 10) \times R_c}{(R_b + 10) + R_c} + 50$$

$$R_t = 20 + 3.33 + \frac{(3.33 + 10) \times 3.33}{(3.33 + 10) + 3.33} + 50$$

$$R_t = 75.994 \Omega$$

$$I = \frac{V}{R_t} = \frac{200}{75.994} = 2.63 A$$

$$P = VI = 200 \times 2.63 = 526 W$$

الأسبوع ٤٠

September 9 أيلول

Sat.	27	20	13	06	السبت
Sun.	28	21	14	07	الأحد
Mon.	29	22	15	08	الاثنين
Tue.	30	23	16	09	الثلاثاء
Wed.		24	17	10	الأربعاء
Thu.		25	18	11	الخميس
Fri.		26	19	12	الجمعة

NOTES

ملاحظات